Structure-Based Profile Hidden Markov Model for Kunitz Domain Identification

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Abstract

Structural domain identification is a fundamental task in bioinformatics, particularly for specialized families like the Kunitz-type protease inhibitors. Characterized by a highly stable fold that regulates proteolytic pathways, this domain represents an ideal target for profile-based sequence analysis. In this study, a comprehensive bioinformatics pipeline was developed to build and benchmark a profile Hidden Markov Model (HMM) optimized for the accurate detection of Kunitz domain sequences.

A dataset of 134 structures retrieved from the Protein Data Bank (PDB) was filtered by resolution (≤3.5Å) and sequence length (40–70 residues). Sequence redundancy was subsequently reduced via MMseqs2 clustering, yielding 13 non-redundant representatives. A multiple sequence alignment (MSA) of these canonical structures was generated via structural superposition using PdbeFold, and the resulting alignment was employed to train the profile HMM. Model performance was assessed using a two-fold evaluation scheme, splitting 398 annotated positive sequences from Swiss-Prot and 574,229 true negatives into two independent datasets. Statistical performance evaluation across various E-value cutoffs identified an optimal threshold of 10^{-5} based on the Matthews Correlation Coefficient (MCC). The optimized model demonstrated high discriminative power, achieving an MCC up to 0.9950 in the first set and 0.9848 in the second. At this threshold, the pipeline proved highly reliable, isolating only 5 false negative sequences in the second subset due to non-randomized data partitioning, which are flagged for subsequent biological investigation.

1. Introduction

Kunitz-type proteins represent a widespread family of serine protease inhibitors present throughout all evolutionary lineages, from bacterial cells to plant tissues. The Kunitz-type motif is composed of a short polypeptide chain, typically spanning from 50 to 60 amino acid residues, which exhibits an antiparallel α/β fold stabilized by six highly conserved cysteine residues connected by disulfide bonds in a C1-C6, C2-C4 and C3-C5 arrangement (Fig. 1A). The first two arrangements are central for the preservation of the native conformation whereas the third one stabilizes the two binding domains in the loops that contain the reactive site (Fig. 1B) (de Magalhães et al., 2018).

Bovine pancreatic trypsin inhibitor (BPTI) is the classic representative of this family of proteins and was the first Kunitz-type protease inhibitor described, but the family includes numerous other members. Due to a relatively broad specificity, BPTI can inhibit trypsin as well as chymotrypsin and elastase-like serine enzymes. These Kunitz-type inhibitors exhibit a highly modular architecture, occurring either as single-domain structures or as multi-domain chains containing double, triple, or multiple tandem repeats. This multi-domain setup enables a single polypeptide chain to simultaneously and independently bind several target protease molecules at distinct reactive sites (Ranasinghe and McManus, 2013).

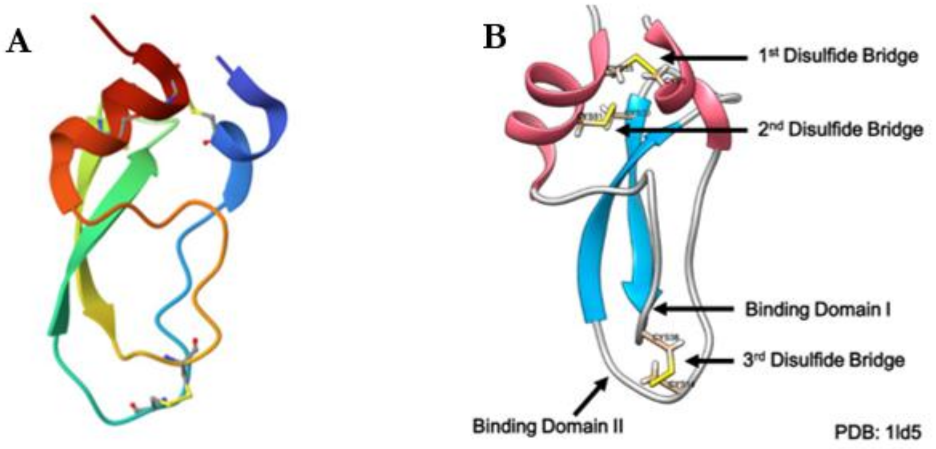


***Fig. 1**: Three-dimensional fold of bovine pancreatic trypsin inhibitor. **(A)** PDB structure of 1LD5 showcasing the canonical Kunitz domain architecture. **(B)** Structural diagram highlighting the invariant cysteine residues and their respective three disulfide bonds highlighted in yellow, while the two antiparallel β-sheets are in blue and a-helix are in red* (de Magalhães et al., 2018).

Hidden Markov Models (HMMs) are statistical frameworks designed to represent a Markov process with hidden, unobservable states. Owing to their capacity to capture dependencies between adjacent states, HMMs are inherently well-suited for sequence-related analyses (Ma et al., 2026).

There is a specific type of HMM that is called **profile Hidden Markov Model**, which is especially useful for representing the profile of a multiple sequence alignment. Unlike general HMMs, profile-HMMs have a strictly linear left-to-right structure that does not contain any cycles. A profile-HMM repetitively uses three types of hidden states, namely, match states M_k , insert states I_k , and delete states D_k , to describe position-specific symbol frequencies, symbol insertions, and symbol deletions, respectively (Yoon, 2009).

Due to their effectiveness in modeling biological sequences, profile-HMMs have been widely adopted for protein family characterization, motif detection, and remote homology identification. Standard software packages, most notably HMMER, provide robust computational tools to build, train, and query these models against large sequence databases. (Yoon, 2009).

Within this framework, the HMMER software suite is employed to both build and train a tailored profile HMM from a curated MSA of Kunitz domain. The aim is to capture the conserved core of the Kunitz-type protease inhibitors while accounting for their evolutionary sequence variations. Once trained, this model serves as a highly sensitive tool to scan extensive databases, enabling the accurate identification and classification of Kunitz domains that standard sequence comparison methods might miss.

2. Materials and methods

2.1 Dataset preparation and Multiple Structural Alignment

The first step of the project involved identifying and selecting high-quality three-dimensional structures of the Kunitz domain from Protein Data Bank (PDB). An advanced search was performed using these criteria:

- Pfam protein family identifier PF00014.
- Data collection resolution was restricted to a maximum of 3.5 Å to ensure high structural quality and reliable atomic coordinates.
- Polymer entity sequence length restricted to a window between 40 and 70 residues to guarantee that only single-domain structures were selected, preventing multi-domain alignment complications.

This preliminary filtering pipeline retrieved an initial cohort of 134 structures.

Following the initial structural retrieval, a custom tabular report was configured to extract essential metadata, specifically selecting the Entity ID, Auth Asym ID (chain), and amino acid sequence for each entry, and the output was saved in JSON format.

This JSON file was then cleaned and processed using a custom Python script. This parsing step successfully unraveled and isolated every individual macromolecular chain within the PDB structures and yielding a total of 238 individual polypeptide chains. During this pipeline, a length filter was strictly applied to retain only sequences between 40 and 80 residues, discarding anomalous fragments or large co-crystallized partner proteins.

To eliminate redundant and highly similar sequences from the dataset, a clustering step was performed using the MMseqs2 web server. The software was configured to group sequences sharing at least 90% sequence identity and 90% alignment coverage. Single-sequence clusters that showed no significant similarity to the main group were discarded. Finally, the first sequence of each remaining cluster was selected as the representative entry. This procedure effectively reduced the dataset to 13 non-redundant representative structures.

To define the common structural core, the identifiers (IDs) of these 13 selected representative sequences were compiled and submitted to the pdbeFold (SSM) online server. This platform was utilized to perform a multiple structural alignment (MSA), allowing for a precise spatial comparison based on the three-dimensional coordinates of the proteins, which yielded an overall Root-Mean-Square Deviation (RMSD) of 0.7316 Å and a global Q-score of 0.5532.

2.2 Profile Hidden Markov Model Generation

Using the hmmbuild tool from HMMER (v.3.4), a Hidden Markov Model (HMM) profile for the Kunitz domain was built. The multiple sequence alignment obtained from the previous step was used as the input to build the mathematical profile which localized the core aligned domain between consensus positions 8 and 63. The model was configured to calculate the emission and transition probabilities for each match state across the aligned positions. To visualize the conservation pattern across the domain, a sequence logo was generated using Skylign, highlighting amino acid preferences at each match state.

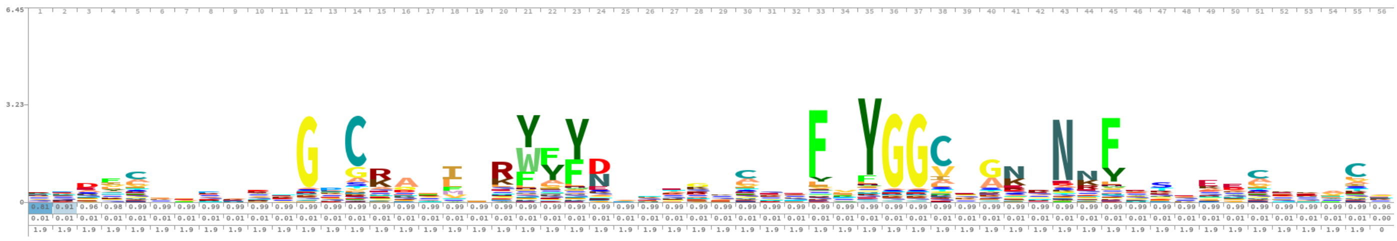


Fig. 2: Sequence logo of the Kunitz domain generated from the profile Hidden Markov Model. *The height of each amino acid letter is proportional to its conservation level (measured in bits) at that specific match state across the 56 consensus positions. The logo shows clearly the 6 conserved cysteine involved in the three disulfide bridges that stabilize the fold.*

2.3 Reference Dataset preparation and HMM model evaluation

To evaluate the predictive power and robustness of the generated Hidden Markov Model (HMM), a statistical validation procedure was performed using the curated Swiss-Prot sequence database (UniProtKB).

Using the UniProt advanced search engine, two independent datasets were extracted:

- Positive Dataset: Composed of all 398 reviewed proteins belonging to the Kunitz domain Pfam family (PF00014).
- Negative Dataset: Composed of the entire sequence space of Swiss-Prot (574,229 proteins), strictly excluding the domain family under investigation.

We used hmmsearch from the HMMER package to test our model against both datasets. The model found matches for all 398 sequences in the positive dataset. However, it found matches for only 6,292 sequences in the negative dataset. This happens because, by default, HMMER automatically filters out sequences that do not show a significant score.

To avoid statistical evaluation bias caused by missing data, the remaining 567,937 negative proteins were manually recovered and reintroduced into the evaluation pool. These unmatched entries were assigned a baseline, synthetic E-value of 100 (exceeding the maximum empirical match score recorded).

The results from both searches were then parsed into classification tables containing the sequence IDs, the E-values, and the labels representing the real biological class (labeled as 1 for true Kunitz positives and 0 for true negatives).

To investigate threshold stability, a balanced two-fold data splitting strategy was implemented to partition the data into two independent evaluation subsets:

- Set 1: Populated with exactly half of the positive matches (199 lines) and half of the total negative data (287,115 lines).
- Set 2: Populated with the remaining half of the positive matches (199 lines) and negative data (287,114 lines).

After partitioning the data, we evaluated the predictive power of our Hidden Markov Model. The main goal was to find the best E-value threshold to accurately separate the true Kunitz domains (positives) from the non-Kunitz proteins (negatives).

To do this, we used a custom Python script which computes the complete Confusion Matrix (TP, TN, FP, FN) along with Overall Accuracy (Q_2), Sensitivity, Specificity and the Matthews Correlation Coefficient (MCC) for each of the 15 different E-value thresholds tested, ranging from 10^{-1} down to 10^{-15}. This grid-search was automated using a bash for loop on the terminal.

We applied this loop independently to both Set 1 and Set 2. By comparing the detailed classification outputs and the metric trends of the two sets, we could check if our optimized threshold was stable, reliable, and not affected by overfitting.

3. Results and discussion

The multiple structural alignment of the 13 non-redundant Kunitz representatives revealed a remarkably high architectural conservation, as shown by an overall RMSD of 0.7316 Å and a global Q-score of 0.5532. This structural consistency allowed the trained profile HMM to define a clean, uninterrupted core domain spanning from consensus position 8 to position 63, effectively filtering out variable terminal regions. The corresponding sequence logo (Fig. 2) clearly highlighted the structural framework of the fold, showing absolute conservation of specific Cysteine residues essential for forming the three intra-domain disulfide bonds. Highly distinct conservation peaks were also visible around the key functional G-C, Y-Y, and E-Y-G-G-C motifs.

During the initial reference database screening, the profile HMM demonstrated excellent sensitivity by successfully assigning matching scores to every single sequence in the Swiss-Prot positive dataset. Regarding the negative database, the model successfully rejected the vast majority of non-related proteins, leaving a total of 567,937 unmatched negative entries that were safely assigned a baseline, synthetic E-value of 100.

The generated HMM model achieved strong discriminatory power while balancing sensitivity and specificity. During the 2-fold cross-testing procedure, multiple E-value cutoffs were tested by assessing the Matthews Correlation Coefficient (MCC) alongside other statistical metrics (Fig. 3).

While Set 1 displayed a perfect mathematical performance plateau (MCC = 1,0000) that stretched across stricter thresholds from 10^{-6} down to 10^{-12}, Set 2 reached its highest performance peak (MCC = 0,9849) at 10^{-3} and maintained a stable, optimal score of 0.9848 down to 10^{-5}.

To guarantee maximum reliability across both independent validation splits, the 10^{-5} threshold was selected as the final cutoff. At this cutoff, the profile HMM demonstrated excellent predictive power, yielding 199 True Positives (TP), 287,113 True Negatives (TN), 2 False Positives (FP), and 0 False Negatives (FN) for Set 1, and 194 True Positives (TP), 287,113 True Negatives (TN), 1 False Positive (FP), and 5 False Negatives (FN) for Set 2. This specific threshold is crucial because tightening the cutoff any further (such as moving from 10^{-5} to 10^{-6}) causes a progressive drop in the performance of Set 2, where the MCC steadily decreases from 0.9822 down to 0.8652 at 10^{-15}.

This drop occurs because Set 2 contains more challenging outlier entries, making the threshold stricter forces the model to miss these atypical entries, transforming them into false negatives. On the other hand, choosing a looser threshold (such as 10^{-1} or 10^{-2}) would compromise specificity by allowing non-related proteins from the negative database to bypass the filter as false positives.

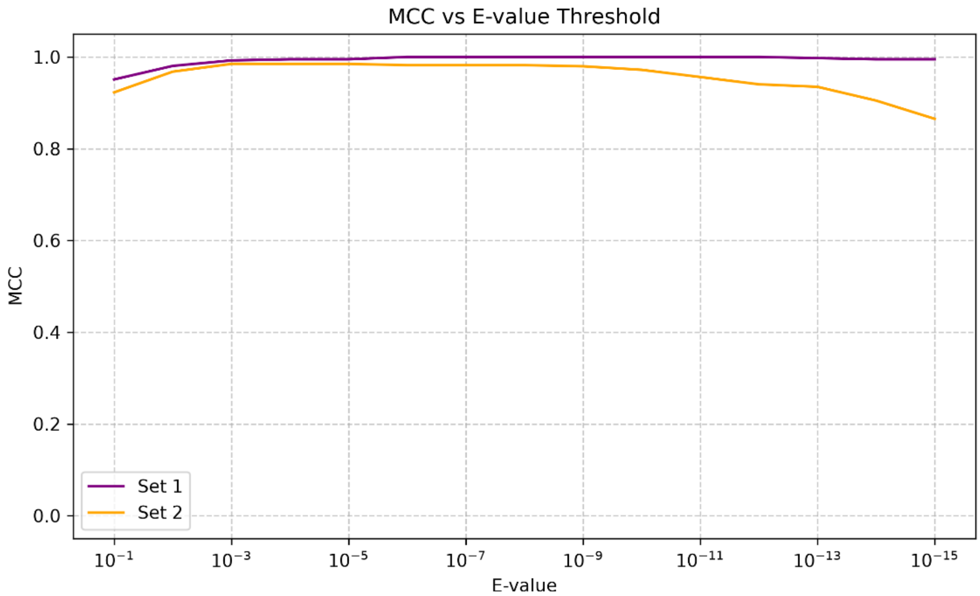
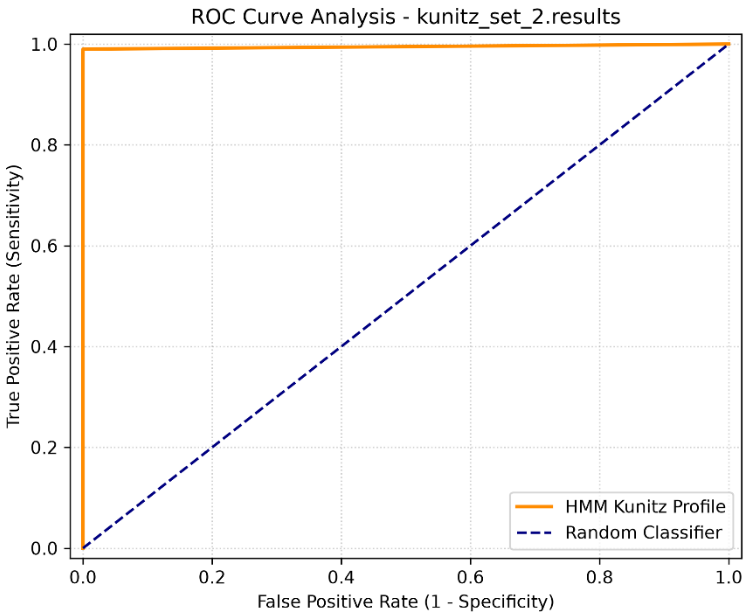


Fig. 3*: *MCC vs E-value threshold performance analysis. *Evaluation of the Matthews Correlation Coefficient (MCC) across different E-value cutoffs ranging from* 10^{-1} *down to* 10^{-15}*. The plot compares the model performance on Set 1 (purple line) and Set 2 (orange line). Both datasets show a stable, optimal performance plateau around the* 10^{-5} *threshold, which effectively isolates true positive Kunitz domains while minimizing classification errors. Stricter cutoffs beyond* 10^{-9} *lead to a progressive drop in performance specifically for Set 2 due to the presence of more challenging outlier sequences.*



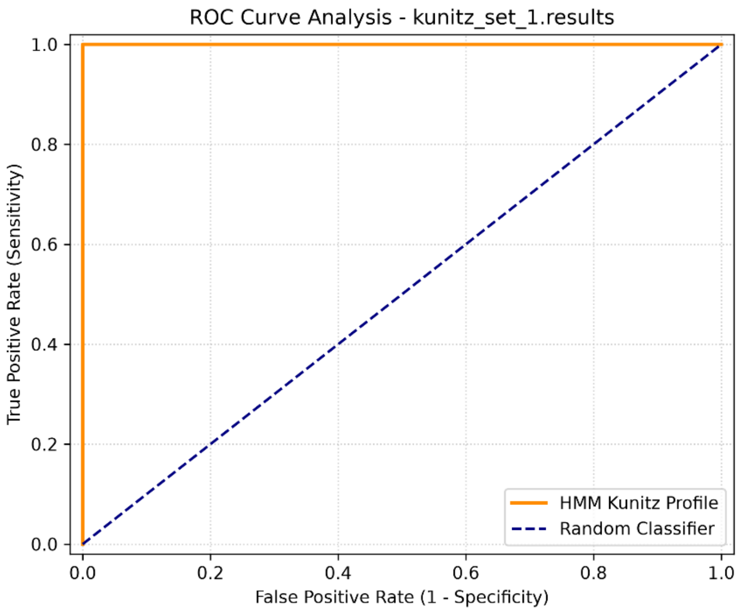


***Fig. 4**: ROC curves The profile HMM achieves a perfect AUC of 1.00 on Set 1 and a near-perfect AUC close to 1.00 on Set 2, indicating exceptional classification performance across varying E-value thresholds for both splits.*

4. Conclusion

In this project, we successfully developed a reliable and sensitive profile Hidden Markov Model tailored for the identification of Kunitz domains. By using high-quality structures from the Protein Data Bank and removing sequence redundancy, the model effectively captured the core structural features and conserved motifs characteristic of this protein family.

The validation against the Swiss-Prot database confirmed the high accuracy of the model. Through a rigorous cross-validation procedure, an E-value threshold of 10^{-5} was identified as the optimal cutoff. This specific threshold provides an excellent balance, maximizing the detection of true Kunitz proteins while successfully minimizing false positives.

In conclusion, this pipeline represents a practical and efficient computational tool for annotating Kunitz-type inhibitors in large biological datasets. The high reliability of the model makes it a solid framework for future sequence analysis and for deeper investigation into atypical family variants.

5. References

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